

Alexey Ananyevskiy

Curriculum Vitae, May 2026

✉ alseang@gmail.com

🌐 www.aananyevskiy.github.io

Employment

- 2022– **Heisenberg Position**, Ludwig-Maximilians-Universität München, Munich, Germany
- 2017–2022 **Deputy Director**, EIMI, St. Petersburg, Russia
- 2017–2022 **Senior Researcher**, PDMI RAS, St. Petersburg, Russia
- 2017–2019 **Senior Researcher**, Chebyshev Laboratory at SPbU, St. Petersburg, Russia
- 2015–2019 **Deputy Head**, Chebyshev Laboratory at SPbU, St. Petersburg, Russia
- 2013–2017 **Postdoc**, Chebyshev Laboratory at SPbU, St. Petersburg, Russia
- 2010–2013 **Lecturer**, Algebra Department at SPbU, St. Petersburg, Russia

Education

- 2010–2013 **Ph.D. in Mathematics**, *SPbU, St. Petersburg, Russia*
Thesis: "On the algebraic K-theory of some varieties and related problems"
Doctoral advisor: Ivan Panin
- 2005–2010 **Specialist in Mathematics**, *SPbU, St. Petersburg, Russia*
Thesis: "Generalization of Colliot-Thélène and Sansuc theorem on principal T-bundles"

Research visits

- Oct – Nov 2026 SLMATH, Berkeley, USA (planned)
- May – Jun 2022 Isaac Newton Institute, Cambridge, UK (online due to travel restrictions)
- Mar 2020 Isaac Newton Institute, Cambridge, UK; suspended due to the COVID-19 pandemic
- Oct 2018 University of Oslo, Oslo, Norway
- Feb – Mar 2018 University of Oslo, Oslo, Norway
- Jan – Apr 2017 Mittag-Leffler Institute, Stockholm, Sweden
- Nov 2016 Universität Osnabrück, Osnabrück, Germany
- Jan – Apr 2015 Institute for Advanced Study, Princeton, USA
- Jun 2014 Universität Duisburg-Essen, Essen, Germany
- May 2013 Fields Institute, Toronto, Canada
- Jul – Aug 2011 Universität Bielefeld, Bielefeld, Germany
- Jun – Jul 2009 Universität Bielefeld, Bielefeld, Germany

Areas of specialization

Motivic Homotopy Theory, Algebraic K-theory, Homogeneous Varieties, Algebraic Geometry

Publications

- 22. Alexey Ananyevskiy, Nikita Geldhauser, *Chow rings of quasi-split geometrically almost simple algebraic groups*, <https://arxiv.org/abs/2408.09390>

21. Alexey Ananyevskiy, Marc Levine, *Combing a hedgehog over a field*, Algebra and Number Theory 19:12 (2025), 2409–2431, <https://doi.org/10.2140/ant.2025.19.2409>
20. Alexey Ananyevskiy, Elden Elmanto, Oliver Röndigs, Maria Yakerson, *On the motivic Adams conjecture*, Journal of Topology 18:2 (2025), e70026, <https://doi.org/10.1112/topo.70026>
19. Alexey Ananyevskiy, *On the $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori in a reductive group*, IMRN, 2022:19 (2022), 15390–15409, <https://doi.org/10.1093/imrn/rnab156>
18. Alexey Ananyevskiy, *Thom isomorphisms in triangulated motivic categories*, Algebraic and Geometric Topology, 21:4 (2021), 2085–2106, <https://doi.org/10.2140/agt.2021.21.2085>
17. Alexey Ananyevskiy, Grigory Garkusha, Ivan Panin, *Cancellation theorem for framed motives of algebraic varieties*, Advances in Mathematics, 383 (2021), 107681, <http://doi.org/10.1016/j.aim.2021.107681>
16. Alexey Ananyevskiy, *SL-oriented cohomology theories*, in *Motivic Homotopy Theory and Refined Enumerative Geometry*, Contemporary Mathematics, 745 (2020), 1–20, <https://doi.org/10.1090/conm/745>
15. Alexey Ananyevskiy, Oliver Röndigs, Paul Arne Østvær, *On very effective hermitian K-theory*, Mathematische Zeitschrift, 294 (2020), 1021–1034, <https://doi.org/10.1007/s00209-019-02302-z>
14. Alexey Ananyevskiy, Alexander Neshitov, *Framed and MW-transfers for homotopy modules*, Selecta Mathematica, 25:26 (2019), <https://doi.org/10.1007/s00029-019-0472-0>
13. Alexey Ananyevskiy, Andrei Druzhinin, *Rigidity for linear framed presheaves and generalized motivic cohomology theories*, Advances in Mathematics, 333 (2018), 423–462, <https://doi.org/10.1016/j.aim.2018.05.013>
12. Alexey Ananyevskiy, *On the zeroth stable $\mathbb{A}^1$ -homotopy group of a smooth curve*, Journal of Pure and Applied Algebra, 222:10 (2018), 3195–3218, <https://doi.org/10.1016/j.jpaa.2017.12.001>
11. Alexey Ananyevskiy, Marc Levine, Ivan Panin, *Witt sheaves and the η -inverted sphere spectrum*, Journal of Topology, 10:2 (2017), 370–385, <https://doi.org/10.1112/topo.12015>
10. Alexey Ananyevskiy, *Stable operations and cooperations in derived Witt theory with rational coefficients*, Annals of K-theory, 2:4 (2017), 517–560, <https://doi.org/10.2140/akt.2017.2.517>
9. Alexey Ananyevskiy, *On the zeroth stable $\mathbb{A}^1$ -homotopy group of a smooth projective variety*, Journal of Mathematical Sciences, 222:4 (2017), 367–369 (translated from Russian), <https://doi.org/10.1007/s10958-017-3306-7>
8. Alexey Ananyevskiy, *On the push-forwards for motivic cohomology theories with invertible stable Hopf element*, Manuscripta Mathematica, 150:1–2 (2016), 21–44, <https://doi.org/10.1007/s00229-015-0799-6>
7. Alexey Ananyevskiy, *On the relation of special linear algebraic cobordism to Witt groups*, Homology, Homotopy and Applications, 18:1 (2016), 205–230, <https://doi.org/10.4310/HHA.2016.v18.n1.a11>
6. Alexey Ananyevskiy, *The special linear version of the projective bundle theorem*, Compositio Mathematica, 151:3 (2015), 461–501, <https://doi.org/10.1112/S0010437X14007702>

5. Alexey Ananyevskiy, Asher Auel, Skip Garibaldi, Kirill Zainoulline, *Exceptional collections of line bundles on projective homogeneous varieties*, *Advances in Mathematics*, 236 (2013), 111-130, <https://doi.org/10.1016/j.aim.2012.12.016>
4. Alexey Ananyevskiy, *Relationship between algebraic MSL-cobordisms and derived Witt groups*, *Doklady Mathematics*, 87:1 (2013), 76-78 (translated from Russian), <http://doi.org/10.1134/S1064562413010286>
3. Alexey Ananyevskiy, *On the algebraic K-theory of some homogeneous varieties*, *Documenta Mathematica*, 17 (2012), 167-193, <https://www.emis.de/journals/DMJDMV/vol-17/07.html>
2. Alexey Ananyevskiy, Nikolai Vavilov, Sergey Sinchuk, *Overgroups of $E(m; R) \otimes E(n; R)$. I. Levels and normalisers*, *St. Petersburg Mathematical Journal*, 23:5 (2012), 819-849 (translated from Russian), <http://doi.org/10.1090/S1061-0022-2012-01219-7>
1. Alexey Ananyevskiy, Nikolai Vavilov, Sergey Sinchuk, *On the description of overgroups of $E(m; R) \otimes E(n; R)$* , *Journal of Mathematical Sciences*, 161:4 (2009), 461-473 (translated from Russian), <https://doi.org/10.1007/s10958-009-9576-y>

Selected talks

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| December 05, 2025 | <i>On semiorthogonal decompositions for twisted flag varieties</i> , <i>Complex algebraic geometry seminar</i> , Università degli Studi di Milano, Milan, Italy |
| September 04, 2025 | <i>On the semiorthogonal decompositions for twisted flag varieties</i> , <i>Workshop Semiorthogonal decompositions for representations of algebraic groups</i> , Universität Bielefeld, Bielefeld, Germany |
| March 17, 2025 | <i>Nowhere vanishing sections of vector bundles revisited</i> , <i>Conference Motivic homotopy theory</i> , Universität Regensburg, Regensburg, Germany |
| October 22, 2024 | <i>Combing a hedgehog over a field</i> , <i>Algebraic geometry seminar</i> , University of Zurich, Zurich, Switzerland |
| January 15, 2024 | <i>Combing a hedgehog over a field</i> , <i>Algebra seminar</i> , Charles University, Prague, Czech Republic |
| November 23, 2023 | <i>Combing a hedgehog over a field</i> , <i>Kolloquium Geometrie und Arithmetik</i> , Johannes Gutenberg Universität Mainz, Mainz, Germany |
| October 19, 2023 | <i>On the motivic Adams conjecture</i> , <i>AG seminar</i> , Universität Regensburg, Regensburg, Germany |
| September 28, 2023 | <i>Non-Vanishing Sections of Vector Bundles and Chern Classes</i> , <i>ESAGA seminar</i> , Universität Duisburg-Essen, Essen, Germany |
| June 22, 2023 | <i>On the existence of non-vanishing sections of vector bundles</i> , <i>Séminaire d'arithmétique à Lyon</i> , ENS de Lyon, Lyon, France |
| May 10, 2023 | <i>Non-Vanishing Sections of Algebraic Vector Bundles and Trivial Chern Classes</i> , <i>Colloquium of Institut für Mathematik</i> , Universität Osnabrück, Osnabrück, Germany |
| November 17, 2022 | <i>On the existence of a nowhere vanishing section for a vector bundle with trivial top Chern class</i> , <i>Oberseminar zur Algebra und Zahlentheorie</i> , Universität Augsburg, Augsburg, Germany |
| October 27, 2022 | <i>On the existence of a nowhere vanishing section for a vector bundle with trivial top Chern class</i> , <i>seminar Motivic algebraic topology</i> , LMU, Munich, Germany |
| July 25 – 27, 2021 | <i>Motivic homotopy theory and vector bundles</i> , <i>summer school Algebra and Geometry</i> , Yaroslavl, Russia |

- May 20, 2021 *An introduction to motivic homotopy theory*, Mini-workshop on Algebraic Geometry, EIMI, St. Petersburg, Russia
- April 29, 2021 *On the $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori*, Motivic Algebraic Topology Seminar, LMU, online
- January 12, 2021 *On the $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori*, Shafarevich Seminar, online
- September 9, 2020 *On the $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori*, Conference *Motivic Geometry*, CAS, Oslo, Norway (moved to online)
- June 20, 2019 *Motivic cellular structure on some homogeneous varieties*, Conference *Number Theory and Algebraic Geometry (In memoriam of Alexei Zykin)*, Independent University of Moscow, Moscow, Russia
- May 7, 2019 *SL-oriented cohomology theories*, Conference *Algebro-geometric and homotopical methods*, Institute Mittag-Leffler, Stockholm, Sweden
- February 19, 2019 *Motives of smooth algebraic varieties*, Shafarevich Seminar, MI RAS, Moscow, Russia
- October 24, 2018 *Non-vanishing sections of algebraic vector bundles and characteristic classes*, Geometry and Topology Seminar, University of Oslo, Oslo, Norway
- June 29, 2018 *On the linear categories of motives*, Workshop *Motivic homotopy groups of spheres III*, Technische Universität Berlin, Berlin, Germany
- May 14, 2018 *SL-oriented cohomology theories*, Workshop *Motivic homotopy theory and refined enumerative geometry*, Universität Duisburg-Essen, Essen, Germany
- March 6, 2018 *Framed and Milnor-Witt transfers for homotopy modules*, Geometry and Topology Seminar, University of Oslo, Oslo, Norway
- February 7, 2017 *Rigidity for framed presheaves and motivic homotopy groups*, Mittag-Leffler Seminar, Institute Mittag-Leffler, Stockholm, Sweden
- November 29, 2016 *Algebraic K-theory of some homogeneous varieties*, Kollegseminar, Universität Osnabrück, Osnabrück, Germany
- November 22, 2016 *Rigidity for motivic stable homotopy sheaves*, Topology Seminar, Universität Osnabrück, Osnabrück, Germany
- June 24, 2016 *On the zeroth stable $\mathbb{A}^1$ -homotopy group of a smooth curve*, Workshop on motivic homotopy groups of spheres II, Universität Duisburg-Essen, Essen, Germany
- February 3, 2016 *Stable operations in derived Witt theory and motivic Serre finiteness*, Workshop *Algebraic Cobordism and Projective Homogeneous Varieties*, Oberwolfach, Germany
- September 25, 2015 *Operations in derived Witt theory*, Minisymposium *Motivic Homotopy Theory and its application to problems in Algebra and Algebraic Geometry*, Hamburg, Germany
- June 20, 2014 *Special linear oriented cohomology theories*, Workshop *Oriented theories and symplectic cobordism*, Universität Duisburg-Essen, Essen, Germany
- September 28, 2013 *Cohomology theories with invertible Hopf element*, Workshop *Quantum and motivic cohomology, Fano varieties and mirror symmetry*, Euler Institute, St. Petersburg, Russia

- May 8, 2013 *SL-oriented cohomology theories*, Spring School and Workshop on Torsors, Motives and Cohomological Invariants, The Fields Institute for Research in Mathematical Sciences, Toronto, Canada
- April 23, 2013 *SL-oriented cohomology theories*, Shafarevich Seminar, MI RAS, Moscow, Russia
- September 20, 2012 *The special linear version of the projective bundle theorem*, Conference *Algebraic groups and related structures*, Euler Institute, St. Petersburg, Russia

Grants, Fellowships and Awards

Grants

- 2023– DFG research grant AN 1545/4-1, project "Equivariant and weak orientations in the motivic homotopy theory", 226.650 euro. Joint with Nikita Geldhauser.
- 2022– DFG Heisenberg Program, grant AN 1545/1-1, project "Motivic homotopy theory and cohomological properties of algebraic varieties", 540.000 euro.
- 2021–2022 "Basis" Foundation grant "Stable motivic homotopy theory, K-theory and generalized motivic cohomology", Principal Investigator, 1.200.000 RUB/year.
- 2020–2022 joint RSF-DFG grant 20-41-04401 "Algebraic bordism spectra: Computations, filtrations, applications", Principal Investigator on the Russian side, 6.000.000 RUB/year on the Russian side. Principal Investigator on the German side is Oliver Röndigs.
- 2014–2015 RFBR grant 14-01-31095-mol_a "*Homotopical methods in algebraic geometry and derived Witt groups*", Principal Investigator, 400.000 RUB/year.
- Investigator RFBR grants 19-01-00513-a, 18-31-20044-mol_a_ved, 16-01-00750-a, 15-01-03034-a, 13-01-00429-a, 12-01-33057-mol_a_ved, 12-01-31100-mol_a, 10-01-00551_a, RSF grant 14-11-00456

Fellowships and Awards

- 2020 Simons Foundation Fellowship to stay at Isaac Newton Institute, Cambridge
- 2018–2020 "Young Russian Mathematics" fellowship, 480.000 RUB/year
- 2017 Fellowship to stay at Mittag-Leffler Institute, Stockholm
- 2015 IAS fellowship to stay at Institute for Advanced Study, Princeton
- 2013–2015 Dynasty foundation fellowship for Russia's young mathematicians, 360.000 RUB/year
- 2013, 2015 Gazprom Neft prize for young mathematicians
- 2012 Young Mathematician prize of the St. Petersburg Mathematical Society
- 2010, 2012 Rokhlin prize for St. Petersburg young mathematicians

Mentorship and Research Direction

- 2025–2026 Anton Lin, MSc student at LMU Munich
- 2022– Egor Zolotarev, PhD student at LMU Munich
- 2021–2022 Tariq Syed, Postdoc at EIMI

Services to the community

Refereeing	Adv. Math., Algebr. Geom. Topol., Ann. Inst. Fourier, Ann. K-theory, Compos. Math., Doc. Math., IMRN, Invent. Math., Izv. Math., J. Algebra, J. Inst. Math. Jussieu, J. Pure Appl. Algebra, J. Topology, Math. Proc. Cambridge Philos. Soc., Math Z., Pacific J. Math., St. Petersburg Math. J., Transform. Groups, MathSciNet
Reviewer	German Research Foundation
Organizing Zoom seminar	Zoom seminar on $\mathbb{A}^1$ -topology, motives and K-theory in 2020-2022 https://indico.eimi.ru/category/12/ .
Organizing conferences	[planned] Motifs and Motives, June 29 – July 03, 2026 at LMU, Munich, Germany. Organizers: Alexey Ananyevskiy, Nikita Geldhauser, Andrei Lavrenov and Maksim Zhykhovich. Webpage Emerging Research in Algebraic Groups, Motives, and K-Theory, September 9-13, 2019 at EIMI, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Anastasia Stavrova and Nikolai Vavilov. Webpage Petersburg motives (dedicated to Ivan Panin's 60th birthday), September 2-6, 2019 at EIMI, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Alexander Merkurjev and Serge Yagunov. Webpage Motives in St. Petersburg 2018, September 3-14, 2018 at EIMI, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Grigory Garkusha and Ivan Panin. Webpage Moscow – St. Petersburg algebraic geometry symposium for young mathematicians, May 10-11, 2018 at Chebyshev Laboratory, SPbU, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Valery Gritsenko, Peter Zograf. Webpage